Dino v0.1 — Build Instructions and Bill of Materials

Objective

Upgrade Dino v0.01 from a simple logic sequencer to a functional 8-bit CPU core capable of performing arithmetic operations and register-controlled logic execution.

Combined Bill of Materials (Dino v0.01 + v0.1)

Base Components (Dino v0.01)

Qty	Part	Description
2	SN74LS590N	8-bit binary counters for 16-bit address
1	MB81256-10	32KB SRAM for program/data memory
1	555 Timer	Clock generation
1	SPDT Switch	LOAD/RUN mode selector
2	Pushbuttons	Manual WRITE and CLOCK control
1	8-bit DIP Switch	Byte input during LOAD mode
8	LEDs	Output display for memory byte
8	10kΩ Resistors	Pull-downs for DIP switch
8	220–470Ω Resistors	LED current limiting
1	Breadboard	Prototyping platform
-	Jumper Wires	Wiring interconnects
1	5V Power Supply	Logic and memory voltage source

Phase 2 Additions (Dino v0.1)

Qty	Part	Description
1	74F382N	8-bit ALU (2x 4-bit sections)
2	SN74LS273N	8-bit D-type registers (Register A and B)
1	DM74LS374N	Output latch for ALU result
8	LEDs	Output display for ALU result
8	220–470Ω Resistors	LED current limiting for ALU result display

Optional for flag storage: | Qty | Part | Description | |—–|—————————-|———————————————————————-| | 1 | 74LS74 | D flip-flops for flag capture (Zero/Carry) |

Build Steps

Step 1: Instruction Sequencer Setup (Dino v0.01)

1. Chain 2× SN74LS590N to form 16-bit Program Counter
2. Connect 555 timer (or manual clock button) to CP0 of lower counter
3. Connect RC of lower counter to CP0 of upper counter
4. Route Q0–Q15 from counters to A0–A15 on SRAM
5. Tie /OE on SRAM LOW, /WE connected to WRITE button (active LOW)
6. Use DIP switch to set D0–D7 during LOAD mode
7. In RUN mode, disconnect DIP switch and connect D0–D7 to LED bank

Step 2: Memory Loading

1. Set mode switch to LOAD
2. Set DIP switch to desired byte
3. Press WRITE to store byte to current address
4. Press CLOCK to advance to next address
5. Repeat until program is fully entered

Step 3: Output Verification (Dino v0.01)

1. Set mode switch to RUN
2. Press reset if needed to reset counters
3. Start timer or manually step through memory
4. Observe LEDs reflect stored bytes in sequence

Step 4: Register Wiring (Dino v0.1)

1. Connect SN74LS273N to SRAM data bus for parallel loading (Register A)
2. Connect second SN74LS273N for Register B loading
3. Use DIP switch to test loading or use memory bytes as inputs
4. Wire clock lines to both registers for synchronized latching

Step 5: ALU Integration

1. Connect Register A and B outputs to ALU inputs
2. Set ALU control lines manually or via instruction decoding (future step)
3. Connect ALU output to DM74LS374 latch
4. Connect output latch to LED display through resistors

Step 6: Manual Instruction Simulation

1. Manually load `LOAD_A`, `LOAD_B`, `ADD`, and `OUT` equivalent instructions into SRAM
2. Step through execution and observe results

Step 7: Define Instruction Set (Preview)

Opcode	Mnemonic	Description
0x01	LOAD_A xx	Load Register A with value
0x02	LOAD_B xx	Load Register B with value
0x03	ADD	A = A + B
0x04	STORE_A xx	Store A to memory[xx]
0x05	OUT	Display A on ALU result LED bank

Completion Criteria

- Successfully execute 3–8 byte programs
- Observe correct ALU operations
- Use LED output to verify result
- Ready to expand to instruction decoder and flag-controlled branching